Database Schema Reference (Hazelnut)

1) `survey.surveys`

Purpose: Track survey versions by provider.

Key columns:

- `id uuid` (PK)
- `provider_id uuid` (FK → `survey.survey_providers.id`)
- `provider_survey_id text` (required)
- `version integer` (required)
- `published_at timestamptz`
- `is_active boolean` (required)
- `title text`
- `description text`

Compatibility notes:

- Keep versioning explicit; dashboard ingestion currently selects active/latest survey for mapping.

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2) `survey.questions`

Purpose: Canonical question registry for normalized answers.

Key columns:

- `id uuid` (PK)
- `survey_id uuid` (FK → `survey.surveys.id`)
- `code text` (UNIQUE; canonical identifier used in mappings)
- `type` (user-defined enum/domain; required)
- `is_first_year_only boolean`
- `provider_question_id text`

Compatibility notes:

- `code` must be stable across imports/years when semantic meaning is unchanged.
- `type` must align with value handling in ingestion and KPI logic.

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3) `survey.question_rows`

Purpose: Canonical row metadata for matrix/row-based questions.

Key columns:

- `id uuid` (PK)
- `question_id uuid` (FK → `survey.questions.id`)
- `code text`
- `label text`
- `has_other_text boolean`

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4) `survey.question_columns`

Purpose: Canonical column metadata for matrix/column-based questions.

Key columns:

- `id uuid` (PK)
- `question_id uuid` (FK → `survey.questions.id`)

- `code text`
- `label text`

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5) `survey.question_options`

Purpose: Canonical option metadata for single/multi-select questions.

Key columns:

- `id uuid` (PK)
- `question_id uuid` (FK → `survey.questions.id`)
- `code text`
- `label text`
- `has_other_text boolean`

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6) `survey.provider_mapping`

Purpose: Map provider-native field signatures to canonical question structures.

Key columns:

- `id uuid` (PK)
- Provider-side identifiers:
  - `provider_qid text` (required)
  - `provider_row_code text`
  - `provider_column_code text`
  - `provider_suffix text`
  - `provider_option_code text`
- Canonical targets:
  - `question_code text` (FK → `survey.questions.code`)
  - `row_code text`
  - `row_label text`
  - `column_code text`
  - `column_label text`
  - `option_code text`
  - `option_label text`

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7) `survey.respondents`

Purpose: Person/entity identity participating in surveys.

Key columns:

- `id uuid` (PK)
- `auth_user_id uuid` (FK → `auth.users.id`, nullable)
- `created_at timestamptz`
- `email text`
- `state text`
- `region text`

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8) `survey.farms`

Purpose: Farm-level entity associated with respondent.

Key columns:

- `id uuid` (PK)
- `respondent_id uuid` (FK → `survey.respondents.id`)
- `name text` (required)
- `street text` (required)
- `city text` (required)
- `postcode text` (required)
- `state text` (required)
- `region text` (required)
- `created_at timestamptz`

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9) `survey.survey_responses`

Purpose: One survey submission instance per respondent/farm/year context.

Key columns:

- `id uuid` (PK)
- `respondent_id uuid` (FK → `survey.respondents.id`)
- `survey_id uuid` (FK → `survey.surveys.id`)
- `farm_id uuid` (FK → `survey.farms.id`)
- `year integer` (required)
- `provider_response_id text` (required)
- `submitted_at timestamptz`
- `status text`

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10) `survey.answers`

Purpose: Canonical normalized answer facts used by analytics/KPI.

Key columns:

- `id uuid` (PK)
- `response_id uuid` (FK → `survey.survey_responses.id`)
- `question_id uuid` (FK → `survey.questions.id`)
- `row_id uuid` (FK → `survey.question_rows.id`, nullable)
- `column_id uuid` (FK → `survey.question_columns.id`, nullable)
- `option_id uuid` (FK → `survey.question_options.id`, nullable)
- `value_number numeric`
- `other_text text`
- `created_at timestamptz`

Compatibility notes:

- At least one typed value is typically populated based on question type.
- Multi-select may generate multiple rows for one response/question via different `option_id`.

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11) `survey.raw_responses`

Purpose: Raw response payload landing table, idempotent by provider response ID.

Key columns:

- `id uuid` (PK)

- `provider_response_id text` (UNIQUE, required)
- `ingested_at timestamptz` (required)
- `payload jsonb` (required)

Compatibility notes:

- Keep original source payload shape for lineage and replay.

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12) `survey.raw_response_fields`

Purpose: Parsed per-field staging table derived from raw payload values/labels.

Key columns:

- `id uuid` (PK)
- `provider_response_id text` (required)
- `qid text` (required)
- `field_key text` (required)
- `row_code text`
- `suffix text`
- `value_json jsonb`
- `label_json jsonb`
- `created_at timestamptz`

Compatibility notes:

- This table is an active transformation stage used before applying provider mapping.
- It supports idempotent re-normalization per `provider_response_id`.